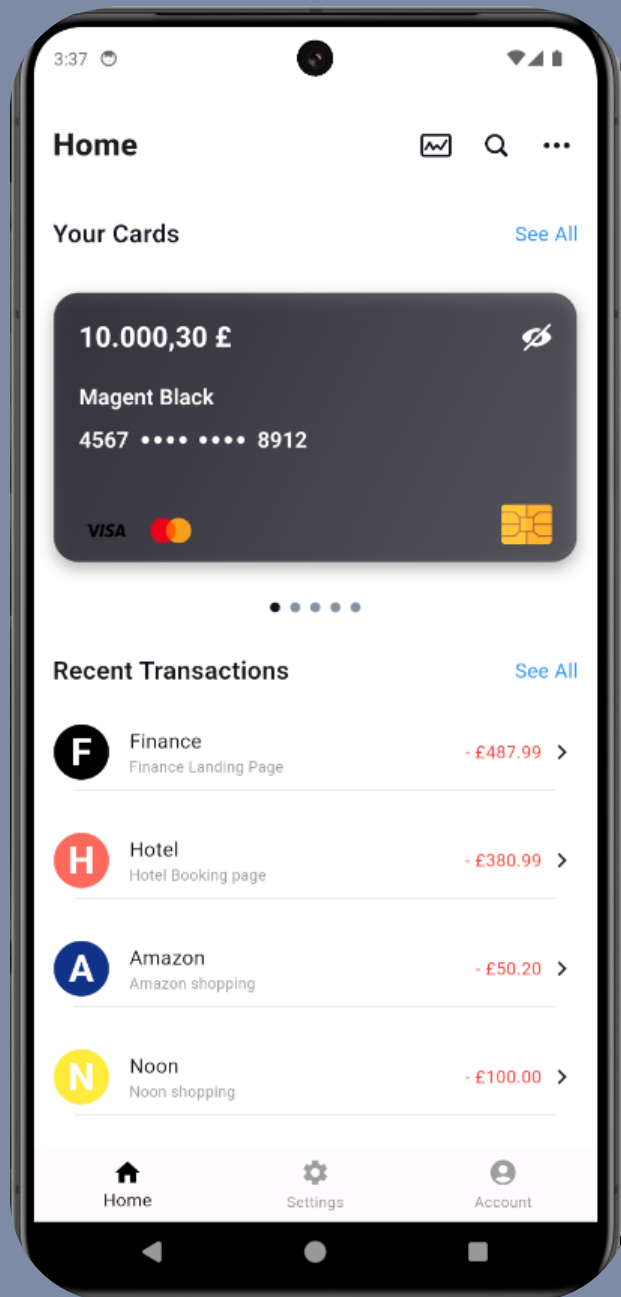




BANKING APP

BY:
Ahmed Mostafa Hussein

Topics:

- 1- Software Requirements Specification (SRS)**
- 2- Use Case Diagram**
- 3- Sequence Diagram**
- 4- Class Diagram**
- 5- Data Flow Diagram**
- 6- Object Diagram**
- 7- Component Diagram**
- 8- State Diagram**
- 9- Project Timing Plan**
- 10- Some Images for Application**

1- Software Requirements Specification (SRS):

1. Introduction

The banking app is a mobile application designed to provide users with convenient access to banking services, including viewing account details, managing transactions, and configuring app settings. This document outlines the functional and non-functional requirements of the banking app, as well as any constraints and assumptions made during the development process.

2. Functional Requirements

2.1 User Management

- **Requirement 1:** The system shall allow users to create an account with a valid email address and password.
- **Requirement 2:** The system shall enforce password complexity requirements (e.g., minimum length, alphanumeric characters) during account creation.
- **Requirement 3:** The system shall allow existing users to log in securely using their registered email address and password.
- **Requirement 4:** The system shall provide a password reset functionality for users who forget their passwords, utilizing email verification for identity confirmation.

2.2 Home Screen

- **Requirement 5:** Upon successful authentication, the app shall display a home screen featuring a bottom navigation bar for seamless navigation between different sections.
- **Requirement 6:** The home screen shall provide quick access to account summary information, including current balance and recent transactions.
- **Requirement 7:** The home screen shall display relevant announcements or promotions from the bank to enhance user engagement.

2.3 Transactions

- **Requirement 8:** The app shall include a transactions page where users can view their recent transactions, organized chronologically.
- **Requirement 9:** Each transaction entry shall include details such as transaction date, description, amount, and recipient/sender information.
- **Requirement 10:** Users shall be able to initiate new transactions from the app, including fund transfers between accounts and payments to external recipients.

2.4 Settings

- **Requirement 11:** The app shall provide a settings page where users can configure app preferences and security settings.
- **Requirement 12:** Settings options shall include language preferences, notification settings, biometric authentication preferences (if supported), and account security settings (e.g., PIN code, fingerprint/face recognition).

3. Non-Functional Requirements

3.1 Security

- **Requirement 13:** The app shall implement end-to-end encryption for all user data transmitted over the network to ensure data confidentiality.
- **Requirement 14:** User authentication mechanisms shall use industry-standard encryption protocols and secure token-based authentication to prevent unauthorized access.
- **Requirement 15:** The app shall store sensitive user data (e.g., passwords, transaction details) securely using encryption and data masking techniques.

3.2 Performance

- **Requirement 16:** The app shall load within 3 seconds on average under normal network conditions.
- **Requirement 17:** Transaction processing shall occur in real-time, with confirmation notifications sent to users within 10 seconds of completing a transaction.
- **Requirement 18:** The app shall handle a minimum of 1000 concurrent users without experiencing performance degradation.

3.3 Usability

- **Requirement 19:** The app shall adhere to platform-specific design guidelines (e.g., Material Design for Android, Human Interface Guidelines for iOS) to ensure consistency and familiarity for users.
- **Requirement 20:** User interface elements shall be intuitively organized and labeled, with clear visual cues to guide users through app navigation and interactions.
- **Requirement 21:** The app shall support accessibility features such as screen reader compatibility and adjustable font sizes to accommodate users with disabilities.

3.4 Reliability

- **Requirement 22:** The app shall be available 24/7 with an uptime of at least 99.9% excluding scheduled maintenance windows.

- **Requirement 23:** Error handling mechanisms shall be implemented to gracefully handle unexpected errors, with informative error messages provided to users for troubleshooting purposes.
- **Requirement 24:** Transactional data shall be stored redundantly with regular backups to prevent data loss in the event of system failures.

4. Constraints

- **Constraint 1:** The app shall be developed using Flutter framework to ensure cross-platform compatibility and code reusability.
- **Constraint 2:** The app shall comply with relevant banking regulations, including data protection laws (e.g., GDPR, CCPA) and financial industry standards (e.g., PCI DSS).
- **Constraint 3:** The app shall support a minimum version of Android 6.0 (Marshmallow) and iOS 12 to maximize market reach and device compatibility.

5. Assumptions

- **Assumption 1:** The app assumes a stable internet connection for accessing banking services and transmitting data securely.
- **Assumption 2:** Users are responsible for maintaining the security of their login credentials and personal devices to prevent unauthorized access to their accounts.
- **Assumption 3:** The app assumes compliance with banking policies and procedures established by the financial institution, including transaction limits, fees, and account restrictions.

6. Glossary

- **User:** An individual who interacts with the banking app to access banking services and manage their financial transactions.
- **Transactions:** Financial activities conducted by users through the app, including fund transfers, bill payments, and account inquiries.
- **Settings:** Configuration options available to users for customizing app behavior, preferences, and security settings.

7. Scope

The scope of the banking app project encompasses the development of a mobile application that provides users with convenient access to banking services on their smartphones. The app will include features such as account management, transaction processing, and settings customization. The primary focus is to deliver a user-friendly and secure platform for users to perform financial activities remotely.

In Scope:

- User authentication and account creation
- Home screen with account summary information
- Transactions page for viewing and initiating transactions
- Settings page for configuring app preferences and security settings
- Secure transmission and storage of user data
- Compliance with banking regulations and industry standards

Out of Scope:

- Integration with third-party financial institutions or payment gateways
- Complex financial analytics or investment management features
- Physical card management (e.g., ordering new cards, activating/deactivating cards)
- Customer support or chatbot functionality
- Offline mode for transaction processing

8. Tools Used

The development of the banking app will leverage a combination of tools and technologies to facilitate the design, implementation, testing, and deployment phases. The following tools will be utilized throughout the project:

Development Tools:

- **Flutter Framework:** A cross-platform UI toolkit for building natively compiled applications for mobile, web, and desktop from a single codebase.
- **Dart Programming Language:** The programming language used for writing code in the Flutter framework.
- **Visual Studio Code (VS Code):** A lightweight, open-source code editor with support for debugging, syntax highlighting, and code completion.

Design Tools:

- **Draw.io:** An online collaborative design tool used for creating UI/UX wireframes, prototypes, and design mockups.

Version Control:

- **Git:** A distributed version control system for tracking changes in source code during development.

Project Management:

- **GitHub:** A web-based platform for hosting Git repositories and managing project collaboration, issue tracking, and task management.

Continuous Integration/Continuous Deployment (CI/CD):

- **GitHub Actions:** Automated workflows for building, testing, and deploying the app to various platforms (e.g., Google Play Store, Apple App Store).

Testing Frameworks:

- **Flutter Driver:** A framework for writing end-to-end integration tests for Flutter apps.
- **Flutter Test:** A framework for writing unit and widget tests to ensure code quality and functionality.

Analytics and Monitoring:

- **Firebase Analytics:** A tool for tracking user interactions and behavior within the app.
- **Firebase Crashlytics:** A tool for monitoring app stability and detecting crashes in real-time.

Security:

- **OAuth 2.0:** An authorization framework used for secure authentication and authorization of user access to resources.
- **SSL/TLS:** Secure Socket Layer/Transport Layer Security protocols for encrypting data transmitted over the network.
- **Firebase Authentication:** A tool for implementing user authentication and identity verification using email/password, phone number, or social login methods.

9.Documentation

Features

- **User Authentication:** Users can securely log in to their accounts using their credentials.
- **Account Management:** Users can view their account balance, transaction history, and perform account-related actions.
- **Transaction Processing:** Users can initiate and process transactions, such as fund transfers and bill payments.
- **Settings Configuration:** Users can customize application settings, including profile details and notification preferences.

Getting Started

- **Installation:** Download the Banking Application from the App Store (iOS) or Google Play Store (Android).
- **Login:** Launch the application and log in using your username and password.

- **Explore:** Navigate through the application to access account information, perform transactions, and configure settings.

User Guide

- **Logging In**
 - Launch the Banking Application on your mobile device.
 - Enter your username and password.
 - Tap the "Login" button to access your account.
- **Account Management**
 - View Balance: Tap on the "Balance" tab to view your account balance.
 - Transaction History: Navigate to the "Transactions" section to view your transaction history.
 - Fund Transfer: Select the "Transfer" option to initiate a fund transfer to another account.
- **Settings Configuration**
 - Profile Settings: Access the "Profile" section to update your personal information.
 - Notification Preferences: Customize your notification settings in the "Notifications" section.

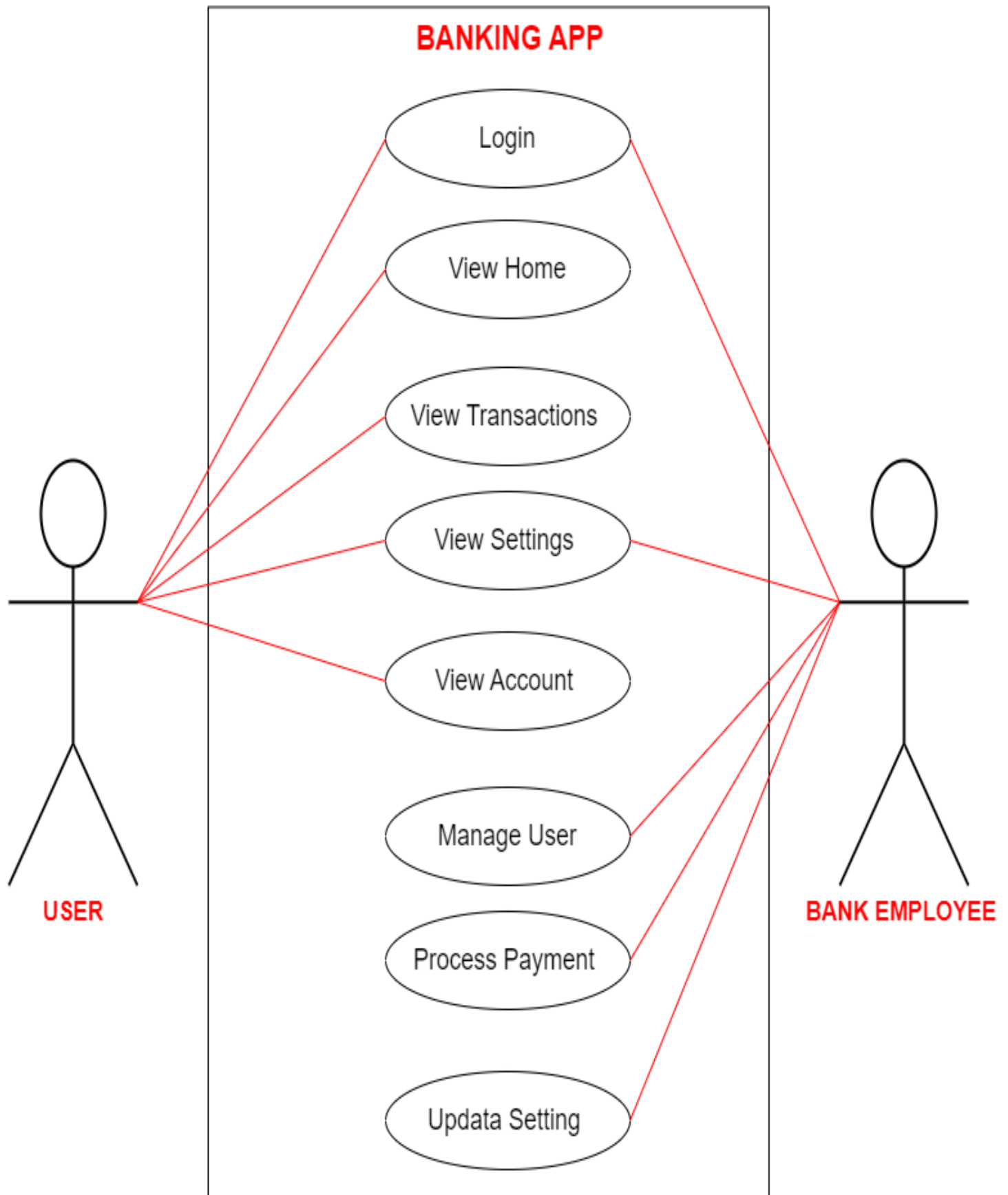
Support

- For assistance or inquiries, please contact our customer support team at support@a7medmostafa7777 or call 01000240832.

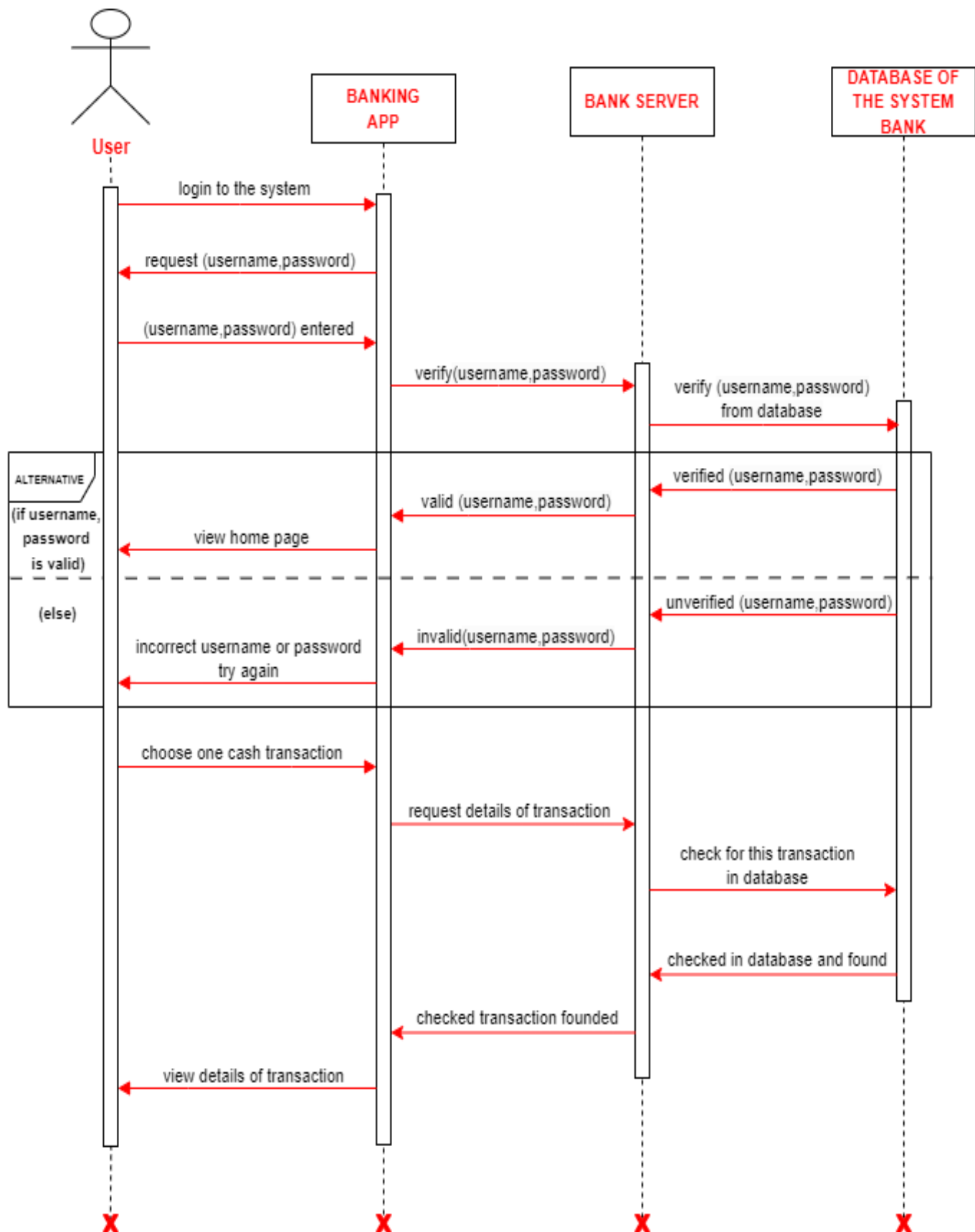
Legal Information

- The Banking Application is provided by [Bank Name]. By using this application, you agree to abide by the terms of service and privacy policy outlined by [Bank Name].

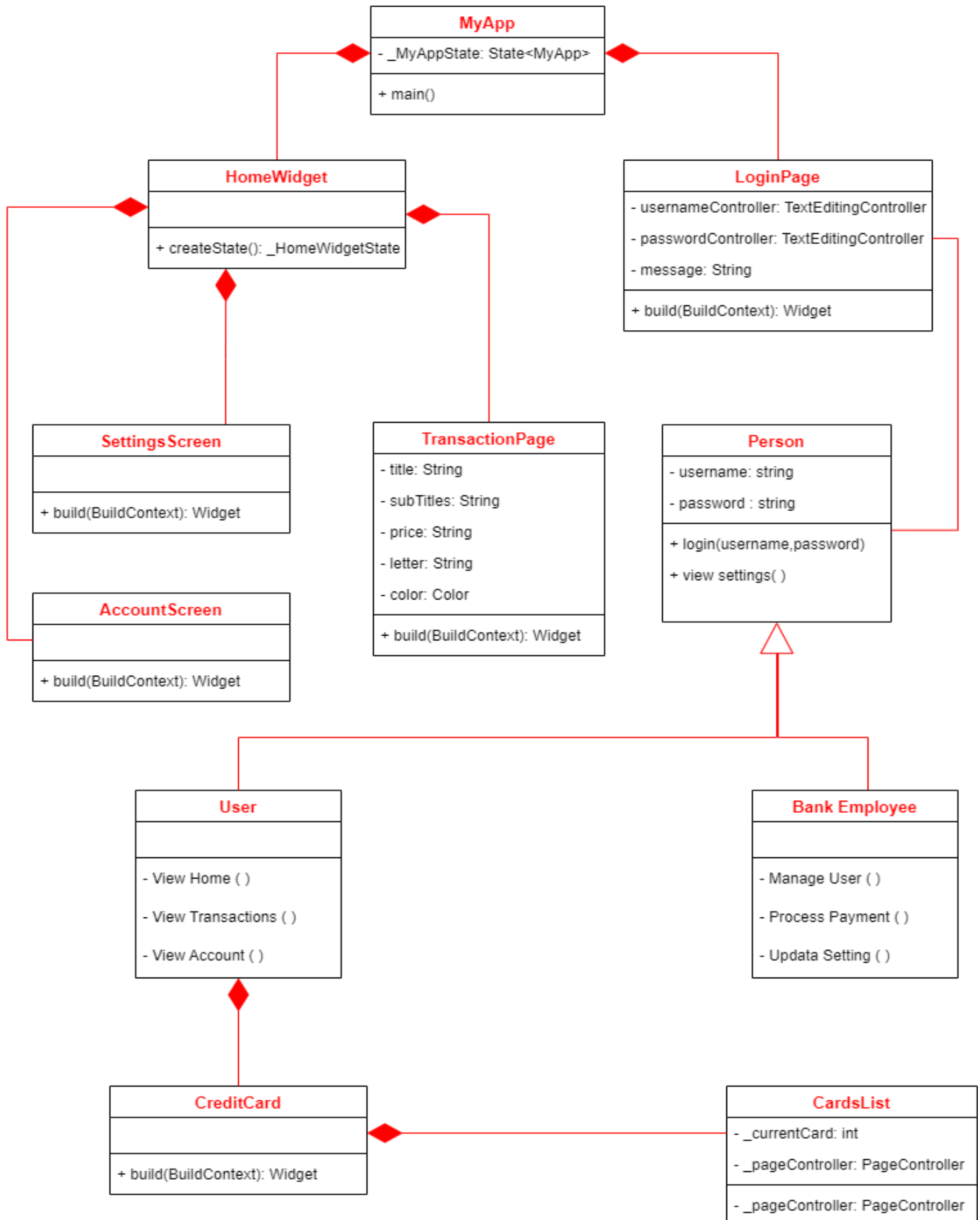
2- Use Case Diagram:



3- Sequence Diagram:

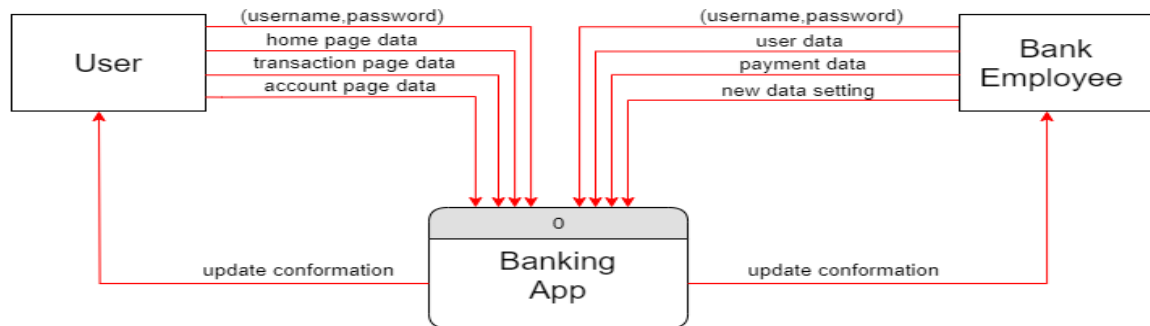


4- Class Diagram:

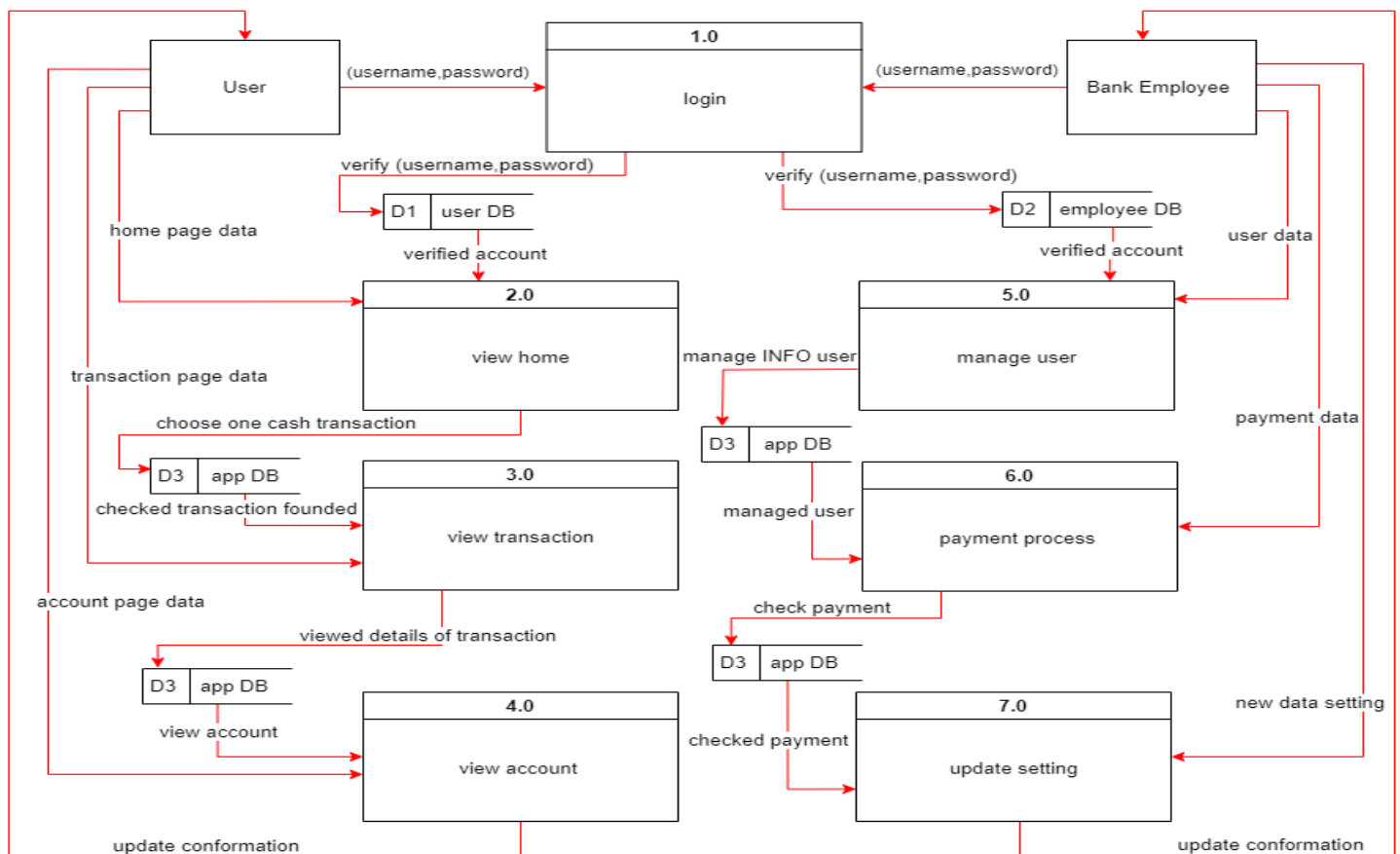


5- Data Flow Diagram:

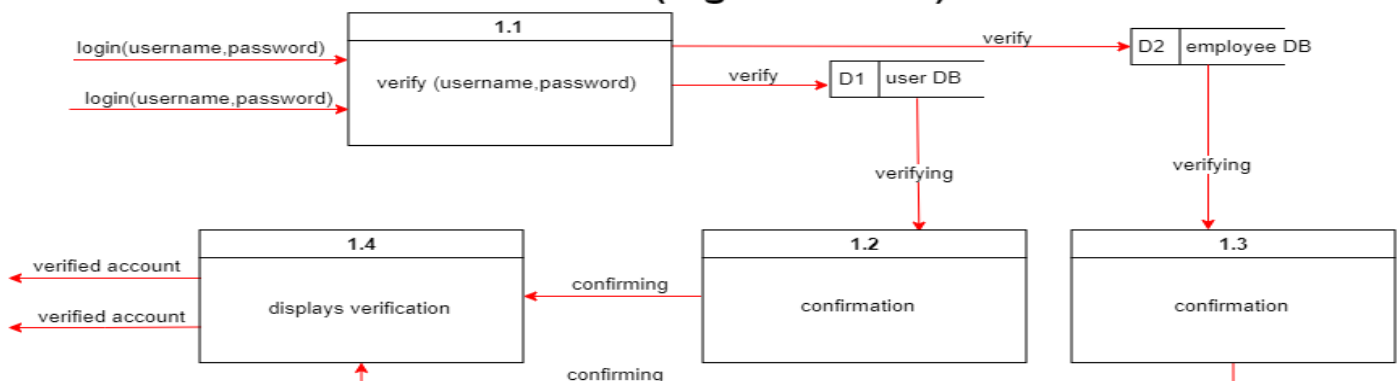
CONTEXT



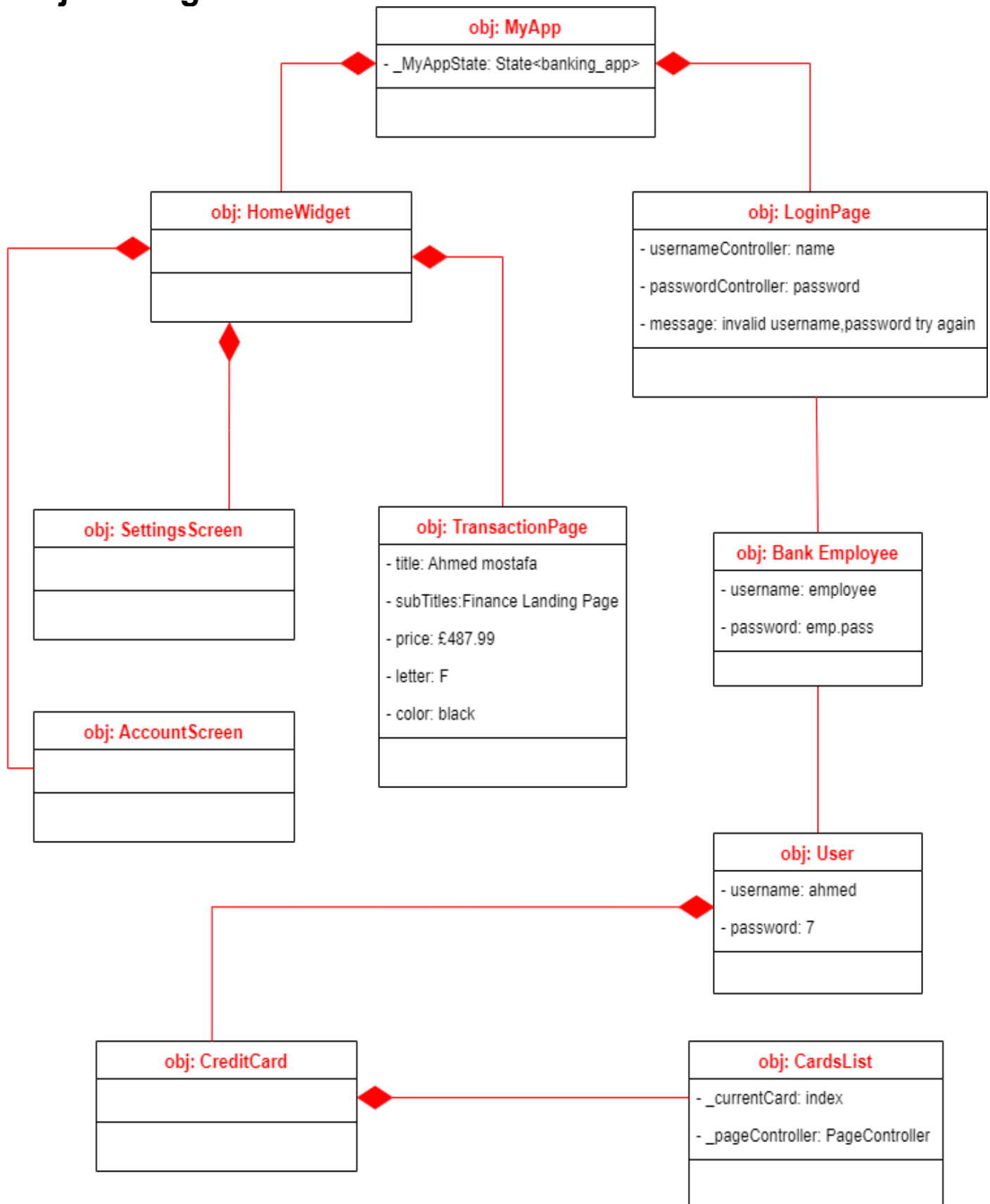
LEVEL 0



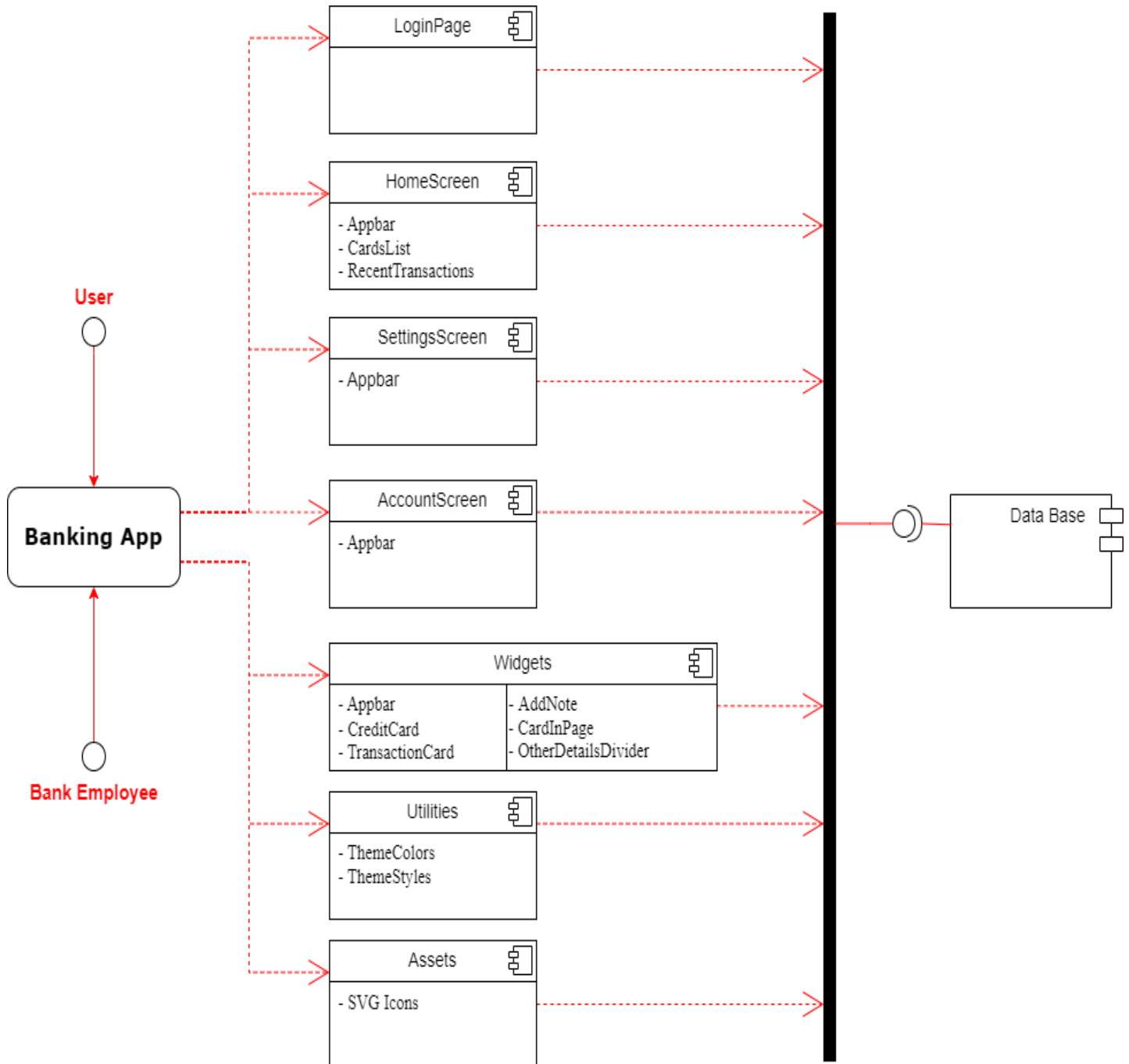
LEVEL 1 (login function)



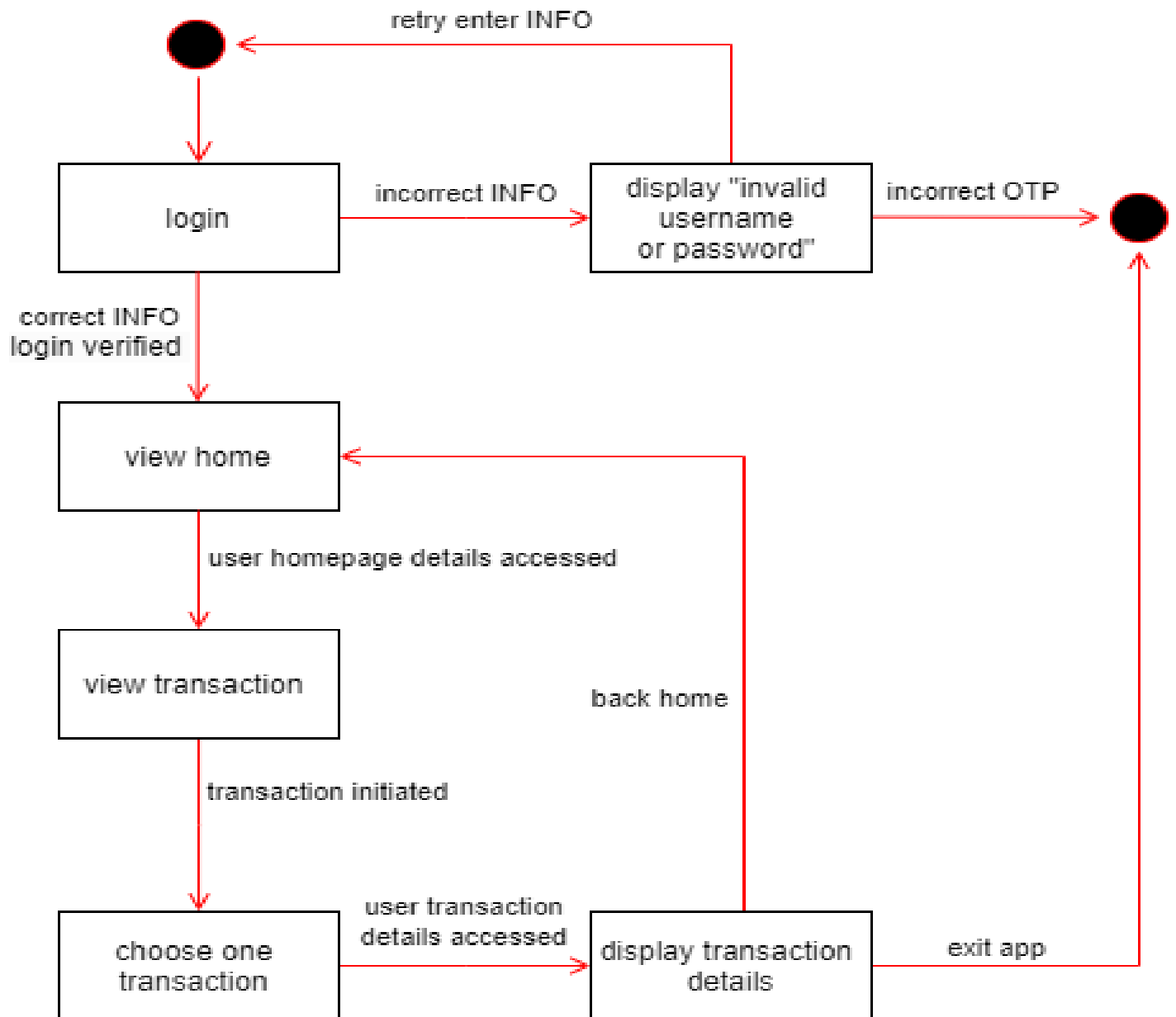
6- Object Diagram:



7- Component Diagram:

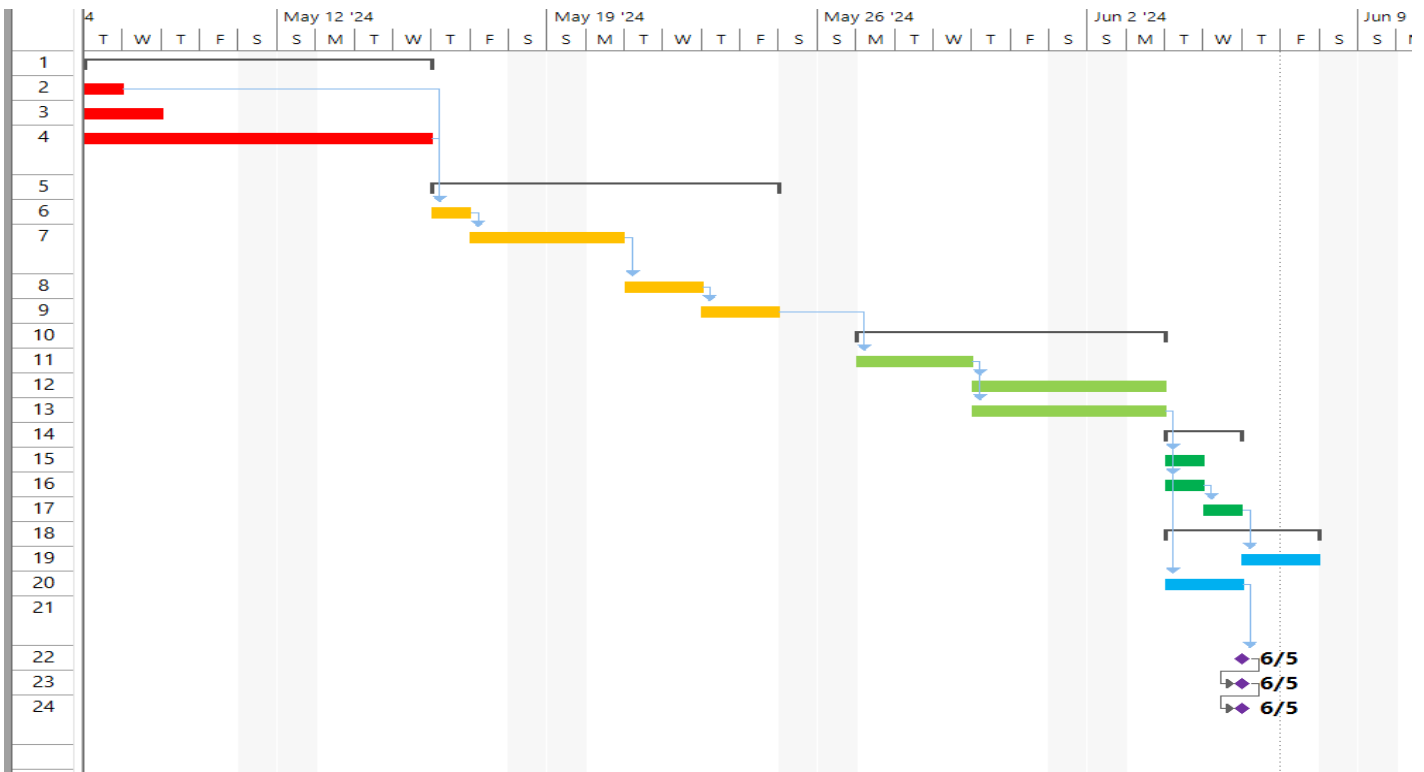


8- State Diagram:



9- Project Timing Plan:

	Task Name	Resource Names	Duration	Start	Finish	Work	Predecessors	% Complete
1	Project Initiation		7 days	Tue 5/7/24 8:00 AM	Wed 5/15/24 5:00 PM	0 hrs		0%
2	scope , objectives		1 day	Tue 5/7/24 8:00 AM	Tue 5/7/24 5:00 PM	0 hrs		0%
3	plan , timeline		2 days	Tue 5/7/24 8:00 AM	Wed 5/8/24 5:00 PM	0 hrs		0%
4	gather requirements		7 days	Tue 5/7/24 8:00 AM	Wed 5/15/24 5:00 PM	0 hrs		0%
5	Design		7 days	Thu 5/16/24 8:00 AM	Fri 5/24/24 5:00 PM	0 hrs		0%
6	architecture		1 day	Thu 5/16/24 8:00 AM	Thu 5/16/24 5:00 PM	0 hrs	2,4	0%
7	wireframes , mockups		2 days	Fri 5/17/24 8:00 AM	Mon 5/20/24 5:00 PM	0 hrs	6	0%
8	database		2 days	Tue 5/21/24 8:00 AM	Wed 5/22/24 5:00 PM	0 hrs	7	0%
9	APIs , data flow		2 days	Thu 5/23/24 8:00 AM	Fri 5/24/24 5:00 PM	0 hrs	8	0%
10	Development		6 days	Mon 5/27/24 8:00 AM	Mon 6/3/24 5:00 PM	0 hrs		0%
11	structure project		3 days	Mon 5/27/24 8:00 AM	Wed 5/29/24 5:00 PM	0 hrs	9	0%
12	interface app		3 days	Thu 5/30/24 8:00 AM	Mon 6/3/24 5:00 PM	0 hrs	11	0%
13	Features		3 days	Thu 5/30/24 8:00 AM	Mon 6/3/24 5:00 PM	0 hrs	11	0%
14	Testing		2 days	Tue 6/4/24 8:00 AM	Wed 6/5/24 5:00 PM	0 hrs		0%
15	components		1 day	Tue 6/4/24 8:00 AM	Tue 6/4/24 5:00 PM	0 hrs	13	0%
16	usability UI/UX		1 day	Tue 6/4/24 8:00 AM	Tue 6/4/24 5:00 PM	0 hrs	13	0%
17	Fix bugs , issues		1 day	Wed 6/5/24 8:00 AM	Wed 6/5/24 5:00 PM	0 hrs	16	0%
18	Deployment		4 days	Tue 6/4/24 8:00 AM	Fri 6/7/24 5:00 PM	0 hrs		0%
19	optimize app		2 days	Thu 6/6/24 8:00 AM	Fri 6/7/24 5:00 PM	0 hrs	17	0%
20	documentation		2 days	Tue 6/4/24 8:00 AM	Wed 6/5/24 5:00 PM	0 hrs	13	0%
21	Maintenance and Updates		0 days	Wed 6/5/24 5:00 PM	Wed 6/5/24 5:00 PM	0 hrs		0%
22	user feedback		0 days	Wed 6/5/24 5:00 PM	Wed 6/5/24 5:00 PM	0 hrs	20	0%
23	issues reported		0 days	Wed 6/5/24 5:00 PM	Wed 6/5/24 5:00 PM	0 hrs	22	0%
24	new features , updates		0 days	Wed 6/5/24 5:00 PM	Wed 6/5/24 5:00 PM	0 hrs	23	0%



10- Some Images for Application:

